OPPERMANN INTERVALS IN $\mathbb{F}_q[t]$: COUNTEREXAMPLES IN CHARACTERISTIC 2 AND A COMPUTER-ASSISTED THEOREM IN ODD CHARACTERISTIC

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ABSTRACT. For a monic $f \in \mathbb{F}_q[t]$ of degree d the Legendre interval $\{f^2 + s : \deg s \le d\}$ splits into the q *Oppermann sub-intervals* $O_{f,c} = \{f^2 + cf + s : \deg s < d\}$, $c \in \mathbb{F}_q$, each of size q^d; for $q = 2$ these are the two halves, the analogue of $(n^2, n^2 + n)$ and $(n^2 + n, (n+1)^2)$. We ask whether every $O_{f,c}$ contains an irreducible polynomial. In characteristic 2 the answer is no, and the failures are explained completely by three Galois-theoretic obstructions: (A) for $d = 1$ the piece $c = 0$ consists of squares; (B) for $d = 2$ we compute the exact number of irreducibles in every Oppermann sub-interval, $q^2/4$, 0, $q(q-2)/4$ or $q(q-1)/2$, by letting Frobenius act on the roots of an additive polynomial through $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$; it vanishes exactly when $q = 4^j$, $a_1^2 + c = 0$, or $q = 2$, $a_1^2 + c = 1$; (C) for $q = 2$, $d = 4$ the block $\{t^8 + s : \deg s \le 3\}$ contains no irreducible, by the Stickelberger–Swan sign of Frobenius, because $8 \mid \deg$ makes the sign depend only on the tail. In odd characteristic every class of the Hayes group of level d is an Oppermann sub-interval. For $d = 2$ we prove the exact count $\pi = (q^2 - 1 - (q - \eta(-1))\eta(2e))/4 > 0$ for every odd q, via a ternary trace form on $\mathbb{F}_{q^4}$; with (B), this settles $d = 2$ for all q. An exact computation with the Hayes explicit formula shows that all of them contain irreducibles for $q = 3$, $d \le 16$; $q = 5$, $d \le 10$; $q = 7$, $d \le 9$; $q = 9$, $d \le 8$; $q = 11$, $d \le 7$; $q = 13$, $d \le 6$. In characteristic 2 the same computation, for $q \in \{2, 4, 8, 16\}$ up to $q^d \le 2^{26}$, finds no empty Oppermann sub-interval beyond (A)–(C). The Weil bound sits at the square-root barrier for these intervals. For $d = 2$ it gives $\Psi \ge q$, and in the empty classes of (B) this is attained exactly, by fourth powers of linear polynomials. For fixed d and large q, the theorem of Bank, Bary-Soroker and Rosenzweig gives nonemptiness in most cases. The one family it misses for $d \ge 3$ lies at $d = 3$ in characteristic 2. Using the tower $\mathbb{F}_q \subset \mathbb{F}_{q^3} \subset \mathbb{F}_{q^6}$, we count every $d = 3$ class there exactly: $q(q^2-1)/6$ for odd k, and at least $q(q-1)^2/6$ for even k. So in characteristic 2 the cases $d \le 3$ are settled for every q, and for every $d \ge 3$ all Oppermann sub-intervals are nonempty once q is large.

1. INTRODUCTION

Oppermann's conjecture [7] asserts that for every $n \ge 2$ both (n^2, n^2+n) and $(n^2+n, (n+1)^2)$ contain a prime. It implies Legendre's conjecture and lies beyond the Riemann hypothesis, which gives primes only in intervals $[x, x + C\sqrt{x}\log x]$. Over $\mathbb{F}_q[t]$, Legendre intervals $I_f = \{f^2 + s : \deg s \le d\}$ (f monic of degree d) were studied in [3]. Since $f^2 + cf$ and $f^2 + ct^d$ differ by a polynomial of degree $< d$, the interval I_f is the disjoint union of the q sub-intervals

$$O_{f,c} = \{f^2 + cf + s : \deg s < d\} = \{f^2 + ct^d + s : \deg s < d\}, \qquad c \in \mathbb{F}_q, \tag{1}$$

each of cardinality $q^d = |f|$, the norm of f, exactly as $(n^2, n^2 + n)$ has length n. For $q = 2$ there are two halves. We call $O_{f,c}$ the *Oppermann sub-intervals* and ask:

Does every $O_{f,c}$ contain an irreducible polynomial?

The answer depends on the characteristic.

Theorem 1.1 (characteristic 2)**.** *Let $q = 2^k$.*

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Code and data: https://github.com/Ruqing1963/oppermann-function-fields.

(A) *For $d = 1$ the sub-interval $O_{f,0} = \{t^2 + s_0 : s_0 \in \mathbb{F}_q\}$, which is the same set for all q polynomials f, consists of squares. Every $O_{f,c}$ with $c \neq 0$ contains exactly $q/2$ irreducibles.*

(B) *For $d = 2$ and $f = t^2 + a_1 t + a_0$, the number of irreducibles in $O_{f,c}$ depends only on $e = a_1^2 + c$ and equals*

	$e = 0$	$e \neq 0$
k even	0	$q^2/4$
k odd	$q(q-1)/2$	$q(q-2)/4$

The table has two zero entries: k even with $e = 0$, and $q(q-2)/4$ at $q = 2$ (where $k = 1$ is odd). Hence $O_{f,c}$ is empty if and only if either k is even and $c = a_1^2$, or $q = 2$ and $c = a_1^2 + 1$. For $q = 2$ the block $e = 0$ is not *empty: it contains $q(q-1)/2 = 1$ irreducible, $t^4 + t + 1$. For every $q \in \{2, 4, 16, 64, \dots\}$ there is thus exactly one empty sub-interval as a set: B_0, or B_1 when $q = 2$. It is represented by q^2 pairs (f, c), one c for each of the q^2 polynomials f. For $q \in \{8, 32, 128, \dots\}$ there is none.*

(C) *For $q = 2$ and $d = 4$ the block $Z_4 = \{t^8 + s : \deg s \leq 3\}$ contains no irreducible polynomial. It equals $O_{f,c}$ for the 8 pairs $f = t^4 + a_2 t^2 + a_1 t + a_0$ (that is, $a_3 = 0$), $c = a_2$.*

Theorem 1.2 (odd characteristic, computer-assisted)**.** *Let q be odd. Every Oppermann sub-interval $O_{f,c}$ contains an irreducible polynomial whenever*

$$q = 3,\ d \leq 16; \qquad q = 5,\ d \leq 10; \qquad q = 7,\ d \leq 9;$$
$$q = 9,\ d \leq 8; \qquad q = 11,\ d \leq 7; \qquad q = 13,\ d \leq 6.$$

Equivalently (Lemma 2.1), every interval $\{P + s : \deg s < d\}$ with P monic of degree $2d$ contains an irreducible polynomial in these ranges.

Theorem 1.3 (characteristic 2, computer-assisted)**.** *For $q = 2$, $d \leq 26$; $q = 4$, $d \leq 13$; $q = 8$, $d \leq 7$; $q = 16$, $d \leq 5$, the only empty Oppermann sub-intervals are those of Theorem 1.1.*

Theorem 1.4 (odd characteristic, $d = 2$)**.** *Let q be odd and let η be the quadratic character of $\mathbb{F}_q$, with $\eta(0) = 0$. For $f = t^2 + a_1 t + a_0$ and $c \in \mathbb{F}_q$, put $e = 2a_0 + c - a_1^2/2$. Then*

$$\pi(O_{f,c}) = \frac{q^2 - 1 - (q - \eta(-1))\,\eta(2e)}{4}.$$

In particular, every Oppermann sub-interval with $d = 2$ contains at least $q(q-1)/4$ irreducibles if $q \equiv 1 \pmod 4$, and at least $(q+1)(q-2)/4$ if $q \equiv 3 \pmod 4$. Together with Theorem 1.1(B), this settles $d = 2$ for every q.

For fixed d and large q, the theorem of Bank, Bary-Soroker and Rosenzweig [1] covers most classes (Corollary 6.2). The one family it misses for $d \geq 3$ lies at $d = 3$ in characteristic 2, and there we can count every class exactly.

Theorem 1.5 (characteristic 2, $d = 3$)**.** *Let $q = 2^k$ and let c be the coefficient of t^3 in an Oppermann sub-interval with $d = 3$. The sub-interval contains exactly $q(q^2 - 1)/6$ irreducibles if k is odd. If k is even, it contains $q(q-1)^2/6 + q\,m(c)$, where $m(c)$ is the number of non-cubes $\nu \in \mathbb{F}_q^\times$ with $\mathrm{Tr}(c/\nu) = 1$; thus $m(0) = 0$, and $(q+2)/6 \leq m(c) \leq q/2$ for $c \neq 0$.*

Corollary 1.6. *For $d \in \{1, 2\}$, the answer is known exactly for every q (Theorems 1.1(A),(B) and 1.4). In characteristic 2 this also holds for $d = 3$ (Theorem 1.5), and no sub-interval is empty there. For every $d \geq 3$ there is $q_0(d)$ such that every Oppermann sub-interval of degree $2d$ over $\mathbb{F}_q$ with $q \geq q_0(d)$ contains an irreducible polynomial.*

For $d \geq 3$, Corollary 6.2 covers every class except those with q even, $d = 3$ and $p_3 = 0$. Theorem 1.5, proved as Theorem 6.4, covers those, and indeed every class with q even and $d = 3$. In characteristic 2 and $d = 2$, the empty sub-intervals for $q = 2, 4, 16, 64, \dots$ show that no such $q_0(2)$ exists there.

The tables above therefore matter mainly for $q < q_0(d)$; $q_0(d)$ is not explicit.

Throughout, a *sub-interval* means the set $O_{f,c}$. Distinct pairs (f,c) can give the same set; counts of sub-intervals are counts of sets, that is, of Hayes classes.

Section 2 identifies the Oppermann sub-intervals with Hayes classes and explains why characteristic 2 is special: the Oppermann classes are concentrated on prefixes close to squares. Section 3 proves Theorem 1.1, and Section 4 proves Theorem 1.4. Section 5 describes the exact computation behind Theorems 1.2 and 1.3. Section 6 explains why the explicit formula alone cannot go further.

2. Oppermann sub-intervals as Hayes classes

For P monic of degree n write $P = t^n + p_{n-1}t^{n-1} + \cdots + p_0$. Two monic polynomials of degree n lie in the same *Hayes class of level l* if they agree in $p_{n-1}, \dots, p_{n-l}$. Each class is a coset $P + \{s : \deg s < n-l\}$, and the map

$$P \mapsto u^n P(1/u) \bmod u^{l+1} = 1 + p_{n-1}u + \cdots + p_{n-l}u^l$$

identifies the classes with the elements of the multiplicative group $G_l = (1 + u\mathbb{F}_q[u])/(1 + u^{l+1}\mathbb{F}_q[u])$ of order q^l. This map is multiplicative: $\mathrm{class}(PQ) = \mathrm{class}(P)\,\mathrm{class}(Q)$ in G_l. The characters of G_l are the short-interval characters of Hayes [4] and Keating–Rudnick [5]. A character has *exact level j* if it factors through G_j but not through G_{j-1}. There are $q^j - q^{j-1}$ characters of exact level j, for $1 \le j \le l$. By (1), $O_{f,c}$ is the Hayes class of level d and degree $n = 2d$ containing $f^2 + ct^d$.

Lemma 2.1 (odd q)**.** *If q is odd, every Hayes class of level d in degree $2d$ is an Oppermann sub-interval, and it equals $O_{f,c}$ for exactly q pairs (f,c).*

Proof. Write $f = \sum_{i=0}^{d} a_i t^i$ with $a_d = 1$. For $1 \le j \le d$ the coefficient of t^{2d-j} in f^2 is

$$[f^2]_{2d-j} = \sum_{\substack{i+i'=2d-j \\ 0\le i,i'\le d}} a_i a_{i'} = 2a_{d-j} + \Phi_j, \qquad \Phi_j = \sum_{\substack{i+i'=2d-j \\ d-j<i,\,i'<d}} a_i a_{i'},$$

where the term $2a_{d-j} = 2a_d a_{d-j}$ comes from $\{i,i'\} = \{d, d-j\}$. The sum Φ_j runs over ordered pairs, so it includes the square $a_{d-j/2}^2$ when j is even. Φ_j depends only on $a_{d-1}, \dots, a_{d-j+1}$; for example $\Phi_1 = 0$, $\Phi_2 = a_{d-1}^2$, $\Phi_3 = 2a_{d-1}a_{d-2}$.

The map $(a_{d-1}, \dots, a_1) \mapsto ([f^2]_{2d-1}, \dots, [f^2]_{d+1})$ is therefore triangular with diagonal entries 2. Since $2 \in \mathbb{F}_q^\times$, it is a bijection $\mathbb{F}_q^{d-1} \to \mathbb{F}_q^{d-1}$: the prefix of a class determines $a_{d-1}, \dots, a_1$ uniquely, by solving $a_{d-j} = ([f^2]_{2d-j} - \Phi_j)/2$ for $j = 1, \dots, d-1$ in turn.

The remaining coefficient of the class is that of t^d in $f^2 + ct^d$, namely $2a_0 + \Phi_d + c$, where Φ_d is already fixed. For each of the q values of a_0 there is exactly one c giving the prescribed value. So every class arises, and from exactly q of the q^{d+1} pairs (f,c). □

So in odd characteristic the Oppermann question at degree $2d$ is the question whether *every* short interval of length $q^d = q^{n/2}$ contains a prime: no class is singled out.

Lemma 2.2 (q even)**.** *Let q be even. The Oppermann sub-intervals are exactly the Hayes classes of level d (in degree $2d$) in which the coefficient of t^j is 0 for every odd j with $d < j < 2d$. The coefficients of t^d and of t^j for even j with $d < j < 2d$ are arbitrary. There are $q^{\lceil d/2 \rceil}$ such classes among the q^d classes of level d, and each of them equals $O_{f,c}$ for exactly $q^{\lfloor d/2 \rfloor + 1}$ pairs (f,c).*

Proof. In characteristic 2, $f^2 = \sum a_i^2 t^{2i}$ has no odd terms, $a \mapsto a^2$ is bijective on $\mathbb{F}_q$, and the coefficient of t^d is shifted by c. This identifies the classes that occur.

For the count, note that $a \mapsto a^2$ is additive. So the map $(f,c) \mapsto \mathrm{prefix}(f^2 + ct^d)$ is a surjective homomorphism of additive groups from $\mathbb{F}_q^{d+1}$, the coefficients $(a_0, \dots, a_{d-1}, c)$, onto $\mathbb{F}_q^{\lceil d/2 \rceil}$. Its fibres therefore all have size $q^{d+1-\lceil d/2 \rceil} = q^{\lfloor d/2 \rfloor + 1}$. □

For example, the empty classes of Theorem 1.1 at $d = 1$, 2 and 4 are represented by q, q^2 and $2^3 = 8$ pairs (f,c) respectively.

Thus in characteristic 2 the Oppermann sub-intervals are the short intervals around the $q^{\lceil d/2 \rceil}$ "near-squares" $f^2 + ct^d$, a set of measure $q^{-\lfloor d/2 \rfloor}$ among all short intervals. Squares and

near-squares carry Galois-theoretic structure, and that structure produces the counterexamples below.

3. Characteristic 2: the three obstructions

Throughout this section $q = 2^k$ and $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}$.

3.1. **(A) $d = 1$.** Here $f = t + a_0$ and $O_{f,c} = \{t^2 + ct + s_0 : s_0 \in \mathbb{F}_q\}$.

For $c = 0$ the members $t^2 + s_0 = (t + \sqrt{s_0})^2$ are squares.

For $c \neq 0$, substitute $t = cy$: $t^2 + ct + s_0 = c^2(y^2 + y + s_0/c^2)$. This is irreducible if and only if $y^2 + y + s_0/c^2$ has no root in $\mathbb{F}_q$, that is, if and only if $\mathrm{Tr}(s_0/c^2) = 1$. Exactly $q/2$ values of s_0 satisfy this.

3.2. **(B) $d = 2$: an additive polynomial and $\mathrm{AGL}_2(\mathbb{F}_2)$.** For $f = t^2 + a_1 t + a_0$ we have $f^2 = t^4 + a_1^2 t^2 + a_0^2$, so

$$O_{f,c} = B_e := \{t^4 + et^2 + \alpha t + \beta : \ \alpha, \beta \in \mathbb{F}_q\}, \qquad e = a_1^2 + c.$$

By Lemma 2.2 the blocks B_e, $e \in \mathbb{F}_q$, are precisely the Oppermann classes for $d = 2$.

Proposition 3.1. *Let $\alpha \neq 0$. Put $L(t) = t^4 + et^2 + \alpha t$ and $R(y) = y^3 + ey + \alpha$.*

1. *$t^4 + et^2 + \alpha t + \beta$ is irreducible for some β if and only if R has exactly one root in $\mathbb{F}_q$.*
2. *If R has exactly one root in $\mathbb{F}_q$, then $t^4 + et^2 + \alpha t + \beta$ is irreducible for exactly $q/2$ values of $\beta \in \mathbb{F}_q$.*

If $\alpha = 0$, every member $t^4 + et^2 + \beta = (t^2 + \sqrt{e}\, t + \sqrt{\beta})^2$ is a square.

Proof. L is $\mathbb{F}_2$-linear and separable ($L' = \alpha \neq 0$), so $V = \ker L \subset \overline{\mathbb{F}}_q$ is a 2-dimensional $\mathbb{F}_2$-space, and $V \setminus \{0\}$ is the set of roots of $L(t)/t = R(t)$. The Frobenius $\varphi(x) = x^q$ acts on V by some $F \in \mathrm{GL}(V) \cong S_3$. Exactly one of three cases holds: $F = 1$ (R has three roots in $\mathbb{F}_q$), F of order 3 (no root), or F a transvection (exactly one root).

Fix β and a root r of $P_\beta = L + \beta$. The roots of P_β form the coset $r + V$. Since $L(\varphi(r)) = \varphi(\beta) = \beta$, we have $\varphi(r) = r + v$ with $v \in V$, and $\varphi(r + w) = r + (Fw + v)$. So Frobenius acts on the roots through the affine map $T(w) = Fw + v$ of $V \cong \mathbb{F}_2^2$, an element of $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$. P_β is irreducible if and only if T is a 4-cycle.

- If $F = 1$, T is a translation and $T^2 = 1$.
- If F has order 3, $F - 1$ is invertible, so T has a fixed point.
- If F is a transvection, $T^2(w) = w + (F+1)v$, so T is a 4-cycle if and only if $(F+1)v \neq 0$, i.e. $v \notin V^F$.

This proves the "only if" part of (1); the "if" part follows from (2), since $q/2 \geq 1$. For (2), let F be a transvection, so $(F+1)V = V^F$ is a line. Replacing r by $r + w$ replaces v by $v + (F+1)w$, so the class $[v] \in V/V^F \cong \mathbb{F}_2$ depends only on β, and $\beta \mapsto [v]$ is additive. Its kernel consists of the β for which some root $r' = r + w$ is fixed by φ, i.e. $\beta \in L(\mathbb{F}_q)$. The map $L : \mathbb{F}_q \to \mathbb{F}_q$ is $\mathbb{F}_2$-linear with kernel $V^F = V \cap \mathbb{F}_q$ of order 2, so $|L(\mathbb{F}_q)| = q/2$ and exactly $q/2$ values of β have $[v] \neq 0$. □

Lemma 3.2. *Let $\alpha \neq 0$. The cubic $R(y) = y^3 + ey + \alpha$ has exactly one root in $\mathbb{F}_q$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$.*

Proof. The roots y_1, y_2, y_3 of R are distinct (R is separable as above), with $y_1 + y_2 + y_3 = 0$ and $y_1 y_2 y_3 = \alpha$. Put $\rho = y_1/y_3 + y_2/y_1 + y_3/y_2$ and $\rho' = y_2/y_3 + y_1/y_2 + y_3/y_1$. Even permutations of the roots fix ρ and odd ones exchange ρ, ρ'; using $y_i + y_j = y_k$,

$$\rho + \rho' = \frac{y_1 + y_2}{y_3} + \frac{y_2 + y_3}{y_1} + \frac{y_3 + y_1}{y_2} = 1.$$

Hence ρ, ρ' are the roots of $z^2 + z + \delta$ with $\delta = \rho\rho' \in \mathbb{F}_q$, and Frobenius acts on the roots of R by an odd permutation (a transposition: exactly one root in $\mathbb{F}_q$) if and only if $\rho \notin \mathbb{F}_q$, i.e.

$\mathrm{Tr}(\delta) = 1$. Expanding,

$$\delta = 3 + \sum_k \frac{y_i y_j}{y_k^2} + \sum_k \frac{y_k^2}{y_i y_j} = 1 + \sum_k \frac{\alpha}{y_k^3} + \frac{1}{\alpha}\sum_k y_k^3.$$

From $y_k^3 = e y_k + \alpha$ we get $\sum_k y_k^3 = \alpha$, so $\delta = \sum_k \alpha/w_k$ with $w_k = y_k^3 = e y_k + \alpha$. Substituting $y = (w+\alpha)/e$ (for $e \neq 0$) into $w = y^3$ shows that the w_k are the roots of $w^3+\alpha w^2+(\alpha^2+e^3)w+\alpha^3$, so $\sum_k 1/w_k = (\alpha^2+e^3)/\alpha^3$ and

$$\delta = 1 + e^3/\alpha^2.$$

For $e = 0$ this also holds directly ($w_k = \alpha$). So $\mathrm{Tr}(\delta) = 1$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$. □

Remark 3.3. δ is Berlekamp's characteristic-2 discriminant [2] of R, and R is the cubic resolvent of the quartic; the proof above is self-contained. For $e = 0$ the lemma says: $y^3 = \alpha$ has exactly one root iff k is odd, which is the familiar fact that cubing is bijective on $\mathbb{F}_q$ iff $3 \nmid q-1$.

Proof of Theorem 1.1(B). By Proposition 3.1, $|B_e \cap \{\text{irreducibles}\}| = \frac{q}{2} \cdot \#\{\alpha \neq 0 : \mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)\}$. If $e = 0$ the condition reads $\mathrm{Tr}(1) = 1$, i.e. k odd: the count is $\frac{q}{2}(q-1)$ for k odd and 0 for k even. If $e \neq 0$, $\alpha \mapsto e^3/\alpha^2$ permutes $\mathbb{F}_q^\times$, and $\#\{x \neq 0 : \mathrm{Tr}(x) = 1\} = q/2$, $\#\{x \neq 0 : \mathrm{Tr}(x) = 0\} = q/2 - 1$; with $\mathrm{Tr}(1) = k \bmod 2$ this gives $q^2/4$ (k even) and $\frac{q}{2}(\frac{q}{2}-1)$ (k odd). The count $q(q-2)/4$ vanishes only for $q = 2$. □

The formula was checked by brute force for $q \leq 64$ (script `char2_d2_theorem.py`): criterion of Lemma 3.2, the $q/2$ count of Proposition 3.1(2), and the four block totals.

3.3. **(C) $q = 2$, $d = 4$: the sign of Frobenius.** The Oppermann classes for $q = 2$, $d = 4$ are the four prefixes $(p_7, p_6, p_5, p_4) = (0, x, 0, y)$ (Lemma 2.2). For $x = y = 0$ the class is the *zero block*

$$Z_4 = \{t^8 + a_3 t^3 + a_2 t^2 + a_1 t + a_0\} \subset \mathbb{F}_2[t],$$

equal to $O_{f,c}$ for $f = t^4 + a_2 t^2 + a_1 t + a_0$ and $c = a_2$. Here $a_3 = 0$ is forced, because the coefficient of t^6 in f^2 is a_3^2; this is why there are $2^3 = 8$ pairs rather than 16.

We use the theorem of Stickelberger [8] in the form of Swan [9, Cor. 1]: *if $\tilde{P} \in \mathbb{Z}[t]$ is monic of degree n and its reduction $P \in \mathbb{F}_2[t]$ is squarefree with r irreducible factors, then $r \equiv n \pmod 2$ if and only if* $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$. Equivalently, the Frobenius permutation of the roots of P is even if and only if $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$. An irreducible P of degree 8 has Frobenius an 8-cycle, an odd permutation, and would need $\mathrm{disc}(\tilde{P}) \equiv 5 \pmod 8$.

Lemma 3.4. *Let $\tilde{P} = t^n + \tilde{g}$ with $\tilde{g} \in \mathbb{Z}[t]$, $\deg \tilde{g} < n$, and $8 \mid n$. Then* $\mathrm{disc}(\tilde{P}) \equiv \mathrm{Res}(\tilde{P}, \tilde{g}') \pmod 8$.

Proof. $\mathrm{disc}(\tilde{P}) = (-1)^{n(n-1)/2}\,\mathrm{Res}(\tilde{P}, \tilde{P}') = \prod_{\tilde{P}(\rho)=0}(n\rho^{n-1} + \tilde{g}'(\rho))$, the sign being $+1$ for $8 \mid n$. Expanding the product, every term containing a factor n is n times a symmetric polynomial with integer coefficients in the roots, hence lies in $8\mathbb{Z}$. □

Proof of Theorem 1.1(C). An irreducible member of Z_4 has no root in $\mathbb{F}_2$, so $a_0 = 1$ and $a_1 + a_2 + a_3 = 1$. This leaves four polynomials. $t^8 + t^2 + 1 = (t^4 + t + 1)^2$ is a square. For the other three choose integer lifts and apply Lemma 3.4:

- $t^8 + t + 1$: $\tilde{g}' = 1$, disc $\equiv 1$;
- $t^8 + t^3 + 1$: $\tilde{g}' = 3t^2$, $\mathrm{Res}(\tilde{P}, 3t^2) = 3^8\big(\prod \rho\big)^2 = 3^8 \equiv 1$;
- $t^8 + t^3 + t^2 + t + 1$, lifted as $t^8 + (t+1)^3$: $\tilde{g}' = 3(t+1)^2$, and $\mathrm{Res}(\tilde{P}, 3(t+1)^2) = 3^8 \prod(\rho+1)^2 = 3^8 \tilde{P}(-1)^2$, with $\tilde{P}(-1) = (-1)^8 + 0^3 = 1$, so disc $\equiv 3^8 \equiv 1$.

Suppose one of these three were irreducible over $\mathbb{F}_2$. Then it would be squarefree with $r = 1$, and Stickelberger–Swan would require $\mathrm{disc}(\tilde{P}) \equiv 5 \pmod 8$. That contradicts $\mathrm{disc}(\tilde{P}) \equiv 1$, so none of them is irreducible. The argument does not need squarefreeness to be checked beforehand. □

Remark 3.5. The mechanism is that $8 \mid \deg P$ kills the leading term of P' modulo 8, so the sign of Frobenius is decided by the tail g alone; with $\deg g \leq 3$ the tail is too short to produce the sign -1 on a polynomial with no linear factor. Table 1 shows the zero blocks $Z_d = \{t^{2d} + s : \deg s < d\}$,

which are Oppermann classes for every d. For $d = 8$ ($8 \mid 2d$ again) the tail has degree ≤ 7, half of the squarefree members already have the sign -1, and the block contains 10 irreducibles. The parallel with (B) is exact: in (B) the obstruction is the quotient $S_4 \to S_3$ (the resolvent cubic), in (C) the quotient $S_8 \to \{\pm 1\}$ (the sign).

TABLE 1. The zero blocks $Z_d = \{t^{2d} + s : \deg s < d\} \subset \mathbb{F}_2[t]$. Columns: the squarefree members; those with disc $\equiv 5 \pmod 8$, i.e. Frobenius odd (integer lift with coefficients $0, 1$); those among them with no root in $\mathbb{F}_2$; and the irreducibles. For $d = 4$ the two members with disc $\equiv 5$ are t^8+t^3+t and $t^8+t^3+t^2+1$, which have the roots 0 and 1. Stickelberger–Swan holds on every squarefree member (script `swan_parity.py`).

d	$\lvert Z_d\rvert$	squarefree	disc $\equiv 5$	disc $\equiv 5$, no root	irreducible
2	4	2	2	1	1
3	8	4	3	1	1
4	16	8	2	**0**	**0**
5	32	16	10	4	4
6	64	32	20	8	6
7	128	64	42	17	9
8	256	128	64	24	10
9	512	256	130	61	28
10	1024	512	280	120	49

4. Odd characteristic, $d = 2$: a ternary trace form

Throughout this section q is odd and η is the quadratic character of $\mathbb{F}_q$, with $\eta(0) = 0$.

4.1. **Reduction to one-parameter blocks.** The level-2 classes in degree 4 are $C_{p_3,p_2} = \{t^4 + p_3t^3 + p_2t^2 + \alpha t + \beta : \alpha, \beta \in \mathbb{F}_q\}$. The translation $t \mapsto t - p_3/4$ is defined because 4 is invertible (in characteristic 3 it is $t \mapsto t - p_3$). It preserves irreducibility and maps C_{p_3,p_2} bijectively onto

$$B_e = \{t^4 + et^2 + \alpha t + \beta : \alpha, \beta \in \mathbb{F}_q\}, \qquad e = p_2 - \tfrac{3}{8}p_3^2.$$

For $O_{f,c}$ with $f = t^2 + a_1t + a_0$ we have $p_3 = 2a_1$ and $p_2 = a_1^2 + 2a_0 + c$, so $e = 2a_0 + c - a_1^2/2$.

4.2. **The count.**

Proposition 4.1. *For every $e \in \mathbb{F}_q$, the number of irreducibles in B_e is*

$$\pi(B_e) = \frac{q^2 - 1 - (q - \eta(-1))\eta(2e)}{4}.$$

Proof. A monic irreducible quartic $P \in B_e$ is the minimal polynomial of exactly four elements $x \in \mathbb{F}_{q^4}$ of degree 4, namely its roots. Write $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_{q^4}/\mathbb{F}_q}$. The coefficient conditions on P translate into conditions on x: $p_3 = -\mathrm{Tr}(x)$, and when $\mathrm{Tr}\, x = 0$,

$$p_2 = e_2(x, x^q, x^{q^2}, x^{q^3}) = \tfrac{1}{2}((\mathrm{Tr}\, x)^2 - \mathrm{Tr}(x^2)) = -\tfrac{1}{2}\mathrm{Tr}(x^2).$$

Hence $4\pi(B_e) = N - M$, where

$$N = \#\{x \in W : Q(x) = -2e\}, \qquad W = \ker \mathrm{Tr}, \quad Q(x) = \mathrm{Tr}(x^2),$$

and M counts the elements of W with $Q(x) = -2e$ that have degree < 4 over $\mathbb{F}_q$, i.e. that lie in $\mathbb{F}_{q^2}$.

The form Q. The bilinear form $(x, y) \mapsto \mathrm{Tr}(xy)$ on $\mathbb{F}_{q^4}$ is nondegenerate. Its Gram determinant on a basis $\omega_1, \ldots, \omega_4$ is $D = \det(\omega_i^{q^j})^2$. Frobenius permutes the columns of the matrix $(\omega_i^{q^j})$ by a 4-cycle, an odd permutation, so $\det(\omega_i^{q^j})$ is not fixed by Frobenius. It therefore lies outside $\mathbb{F}_q$ while its square D lies in $\mathbb{F}_q$, so D is a nonsquare in $\mathbb{F}_q$.

Since $\operatorname{Tr}(1 \cdot x) = \operatorname{Tr} x$, the line $\mathbb{F}_q$ is orthogonal to W, and $\mathbb{F}_{q^4} = \mathbb{F}_q \perp W$. The Gram determinant of $\mathbb{F}_q$ is $\operatorname{Tr}(1) = 4$, a square. So $Q|_W$ is a nondegenerate ternary form whose determinant Δ is a nonsquare.

Counting N. For a nondegenerate quadratic form in 3 variables over $\mathbb{F}_q$ with determinant Δ, the number of solutions of $Q(x) = b$ is $q^2 + q\,\eta(-b\Delta)$ [6, Thm. 6.27]; for $b = 0$ this is q^2. With $b = -2e$ and $\eta(\Delta) = -1$ we get $N = q^2 - q\,\eta(2e)$, also for $e = 0$.

Counting M. An element of $\mathbb{F}_{q^2}$ has $\operatorname{Tr} x = 2\operatorname{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(x)$, so it lies in W if and only if $x^q = -x$. Such x form the line $\delta\mathbb{F}_q$, where $\delta^2 = \nu$ is a nonsquare of $\mathbb{F}_q$. For these x, $x^2 \in \mathbb{F}_q$ and $Q(x) = 2(x^2 + x^{2q}) = 4x^2$, so the condition is $x^2 = -e/2$. The squares of elements $s\delta$ are 0 and the nonsquares of $\mathbb{F}_q$, each nonsquare attained twice. Hence $M = 1$ if $e = 0$, and $M = 1 - \eta(-e/2)$ if $e \neq 0$; in both cases $M = 1 - \eta(-2e)$.

Therefore

$$4\pi(B_e) = q^2 - q\,\eta(2e) - 1 + \eta(-2e) = q^2 - 1 - \big(q - \eta(-1)\big)\eta(2e). \qquad \square$$

Proof of Theorem 1.4. Combine the reduction with Proposition 4.1. The minimum over e is attained at $\eta(2e) = 1$. It equals $(q^2 - 1 - (q-1))/4 = q(q-1)/4$ when $\eta(-1) = 1$, i.e. $q \equiv 1 \pmod 4$, and $(q^2 - 1 - (q+1))/4 = (q+1)(q-2)/4$ when $q \equiv 3 \pmod 4$. Both are positive for every odd q. $\square$

Remark 4.2. Brute-force counts confirm the formula for every block B_e when $q = 3$, 5, 7, 9, 11, 13, 25, 27, 49, 81, and for every class C_{p_3,p_2} when $q \le 13$ (script `odd_d2_formula.py`). They also agree with the minima of Table 2. Characteristic 3 needs no special treatment.

The contrast with characteristic 2 is instructive. There the translation $t \mapsto t + a$ cannot remove p_3, since $4 = 0$. The Oppermann blocks are exactly the classes with $p_3 = 0$, on which $t^4 + et^2 + \alpha t$ is *additive*, and this additivity produces the obstruction of Proposition 3.1. In odd characteristic the trace form $\operatorname{Tr}(x^2)$ has the same shape for every e, and the count never drops below about $q^2/4 - q/4$.

5. The exact computation

5.1. **Method.** Here Λ is the von Mangoldt function: $\Lambda(F) = \deg \varpi$ if $F = \varpi^k$ for a monic irreducible ϖ and some $k \ge 1$, and $\Lambda(F) = 0$ otherwise. For a class x of level l and degree n, let $\Psi(x) = \sum_{F \in x} \Lambda(F)$. The explicit formula [4, 5] reads

$$\Psi(x) = q^{n-l} + q^{-l} \sum_{\chi \neq 1} \overline{\chi}(x)\,\psi_n(\chi), \qquad u\frac{L'(u,\chi)}{L(u,\chi)} = \sum_{m \ge 1} \psi_m(\chi) u^m, \tag{2}$$

where for $\chi \neq 1$ the L-function $L(u,\chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\le l - 1$. Its coefficients are the Fourier transforms on G_l of the class distributions of the monic polynomials of degree $< l$; $\psi_n(\chi)$ follows by the Newton recursion, and one inverse FFT on G_l returns $\Psi(x)$ for all q^l classes at once. This is the algorithm of [3], run here at level $l = d$ and degree $n = 2d$. The number of irreducibles in x is

$$\pi(x) = \frac{1}{n}\big(\Psi(x) - E(x)\big), \qquad E(x) = \sum_{k \ge 2,\ k \mid n} \ \sum_{\deg \varpi = n/k,\ \varpi^k \in x} \frac{n}{k}.$$

The proper prime powers in $E(x)$ are enumerated directly (Ben-Or test in degree $\le d$) and placed by the rule $\operatorname{class}(\varpi^k) = \operatorname{class}(\varpi)^k$ in G_l. Each run checks that every $\pi(x)$ is an integer and that $\sum_x (n\pi(x) + E(x)) = q^n$, and reports the largest distance of the floating-point $\Psi(x)$ from an integer (below 10^{-7} in all runs). As an independent check, a brute-force census that tests every polynomial of degree $2d$ for irreducibility (`verify_pointwise_oppermann.py`) agrees with the FFT counts class by class for all $q^{2d} \le 2^{22}$ (`compare_bruteforce.py`), and in minimum, mean and empty classes for all $q^{2d} \le 2^{26}$.

5.2. **Results.** Table 2 lists, for odd q, the minimum of $\pi(x)$ over all classes (all of which are Oppermann sub-intervals by Lemma 2.1) and its ratio to the expected value $q^d/2d$. Table 3 gives the same for the Oppermann classes in characteristic 2. The full output, all degrees, is in the file
`data/hayes_oppermann_census.txt`.

Proof of Theorems 1.2 and 1.3. Exact computation as described above; the minima are in Tables 2 and 3, and the empty classes in characteristic 2 occur exactly at

$$(q, d) \in \{(2,1), (2,2), (2,4), (4,1), (4,2), (8,1), (16,1), (16,2)\},$$

one class each, which are the classes of Theorem 1.1. □

The normalised minimum $\min_x \pi(x)/(q^d/2d)$ tends to 1 in every column: the worst Oppermann sub-interval is only slightly below average, far from the Weil bound discussed next.

TABLE 2. Odd q: irreducible counts π over all q^d Hayes classes of level d in degree $2d$ (all are Oppermann sub-intervals). Last four degrees for each q; all smaller degrees are in the data file, with minimum ≥ 1 throughout.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
3	13	1 594 323	60 556	61 320.1	0.9875
	14	4 782 969	169 300	170 820.3	0.9911
	15	14 348 907	476 000	478 296.9	0.9952
	16	43 046 721	1 341 030	1 345 210.0	0.9969
5	7	78 125	5 418	5 580.3	0.9709
	8	390 625	23 951	24 414.0	0.9810
	9	1 953 125	107 410	108 506.9	0.9899
	10	9 765 625	486 042	488 281.2	0.9954
7	6	117 649	9 500	9 804.0	0.9690
	7	823 543	57 981	58 824.4	0.9857
	8	5 764 801	358 343	360 300.0	0.9946
	9	40 353 607	2 235 915	2 241 867.0	0.9973
9	5	59 049	5 680	5 904.8	0.9619
	6	531 441	43 728	44 286.7	0.9874
	7	4 782 969	339 993	341 640.6	0.9952
	8	43 046 721	2 684 630	2 690 420.0	0.9978
11	4	14 641	1 735	1 830.0	0.9480
	5	161 051	15 804	16 105.0	0.9813
	6	1 771 561	146 403	147 630.0	0.9917
	7	19 487 171	1 388 036	1 391 940.7	0.9972
13	3	2 197	349	366.0	0.9531
	4	28 561	3 450	3 570.0	0.9664
	5	371 293	36 672	37 129.2	0.9877
	6	4 826 809	400 104	402 234.0	0.9947

TABLE 3. Even q: irreducible counts over the $q^{\lceil d/2 \rceil}$ Oppermann classes (Lemma 2.2). Last four degrees for each q; the only empty classes in the whole range are those of Theorem 1.1.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
2	23	4 096	181 349	182 329.3	0.9945
	24	4 096	347 822	349 412.9	0.9951
	25	8 192	669 169	671 025.2	0.9971
	26	8 192	1 287 576	1 290 408.9	0.9977
4	10	1 024	51 908	52 350.0	0.9901
	11	4 096	189 054	190 603.5	0.9916
	12	4 096	695 868	698 820.5	0.9954
	13	16 384	2 575 566	2 580 963.0	0.9979
8	4	64	440	490.0	0.8594
	5	512	3 204	3 276.0	0.9778
	6	512	21 616	21 840.0	0.9895
	7	4 096	149 148	149 796.0	0.9957
16	2	16	**0**	60.0	**0**
	3	256	600	680.0	0.8789
	4	256	7 616	8 160.0	0.9297
	5	4 096	103 680	104 856.0	0.9888

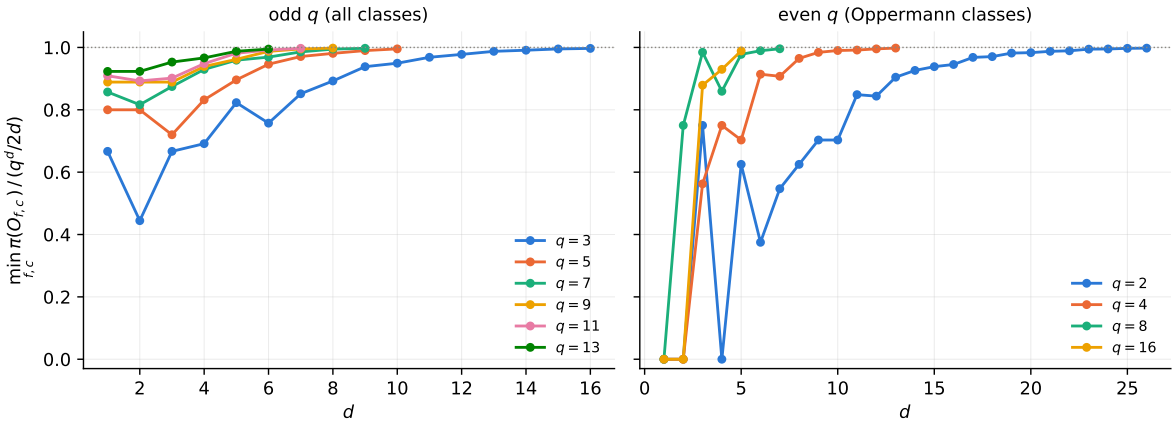


FIGURE 1. Normalised minimum $\min_{f,c} \pi(O_{f,c})/(q^d/2d)$ against d. Left: odd q. Right: even q; the points at 0 are the empty sub-intervals of Theorem 1.1.

6. THE SQUARE-ROOT BARRIER

In (2) with $n = 2d$ and $l = d$ the main term is q^d.

Let χ have exact level j. The monic polynomials of degree $\geq j$ are equidistributed over G_j, so $\deg L(u, \chi) \leq j - 1$. By Weil, every inverse zero γ of $L(u, \chi)$ has $|\gamma| \in \{1, \sqrt{q}\}$, hence $|\psi_{2d}(\chi)| \leq (j-1)q^d$. Characters of level 1 have $L = 1$ and contribute nothing. Multiplying by the factor q^{-d} in (2), which cancels the q^d, and summing over the $q^j - q^{j-1}$ characters of each exact level j gives

$$|\Psi(x) - q^d| \leq \sum_{j=2}^{d} (j-1)(q^j - q^{j-1}) = (d-1)q^d - \sum_{j=1}^{d-1} q^j. \tag{3}$$

For $d \geq 3$ the right-hand side exceeds q^d for every q, so (3) does not even give $\Psi(x) > 0$. This is the exact analogue of the situation over $\mathbb{Z}$: Oppermann's intervals have length $\sqrt{x}$, and the Riemann hypothesis falls short by a factor $\log x$. No degree threshold beyond which $O_{f,c} \neq \emptyset$ follows from Weil's theorem.

For $d = 2$, (3) reads $|\Psi(x) - q^2| \leq q^2 - q$, so $\Psi(x) \geq q > 0$ for every class. This still does not produce an irreducible, because Ψ also counts prime powers. In characteristic 2 the bound is attained, and this pins down every level-2 L-function.

Proposition 6.1 (characteristic 2, $d = 2$)**.** *Let $q = 2^k$, and let $B_e = \{t^4 + et^2 + \alpha t + \beta\}$ as in §3.*

(1) *$E(B_e) = q$ for every $e \in \mathbb{F}_q$.*
(2) *$\Psi(B_0) = q^2 - (-1)^k(q^2 - q)$, and $\Psi(B_e) = q^2 + (-1)^k q$ for $e \neq 0$. In particular, B_0 attains the lower Weil bound q when k is even and the upper bound $2q^2 - q$ when k is odd.*
(3) *Every character χ of exact level 2 has $L(u, \chi) = 1 - \gamma_\chi u$ with $\gamma_\chi^4 = (-1)^k q^2$.*
(4) *For every level-2 class x in degree 4,*

$$\Psi(x) = q^2 - (-1)^k(q^2[x = 1] - q\,[p_3(x) = 0]).$$

In particular, every class with $p_3 \neq 0$ contains exactly $q^2/4$ irreducibles.

Proof. (1) The proper prime powers of degree 4 are ϖ^4 with $\deg \varpi = 1$ and ϖ^2 with $\deg \varpi = 2$.

- $(t+c)^4 = t^4 + c^4$ lies in B_e only for $e = 0$, and then for all q values of c, each with $\Lambda = 1$.
- $(t^2 + b_1 t + b_0)^2 = t^4 + b_1^2 t^2 + b_0^2$ lies in B_e exactly when $b_1^2 = e$. For $e = 0$ this forces $b_1 = 0$, and $t^2 + b_0$ is a square, not irreducible. For $e \neq 0$ we have $b_1 = \sqrt{e} \neq 0$, and by Theorem 1.1(A) the polynomial $t^2 + b_1 t + b_0$ is irreducible for exactly $q/2$ values of b_0, each contributing $\Lambda = 2$.

In both cases $E(B_e) = q$.

(2) We have $\Psi(B_e) = 4\pi(B_e) + q$. Insert the counts of Theorem 1.1(B).

(3) B_0 is the identity of G_2, so $\chi(B_0) = 1$ for every χ. Characters of exact level 1 have $L = 1$ and $\psi_4 = 0$. By (2) and (2),

$$\sum_{\chi \text{ exact level } 2} \psi_4(\chi) = q^2(\Psi(B_0) - q^2) = -(-1)^k(q^2 - q)\,q^2.$$

There are $q^2 - q$ such characters, and each has $|\psi_4(\chi)| \leq q^2$. Equality in the triangle inequality therefore forces $\psi_4(\chi) = -(-1)^k q^2$ for each of them. Then $L(u, \chi)$ cannot be constant, so $L(u, \chi) = 1 - \gamma_\chi u$ and $\psi_4(\chi) = -\gamma_\chi^4$.

(4) By (3) and (2),

$$\Psi(x) = q^2 - (-1)^k \sum_{\chi \text{ exact level } 2} \overline{\chi}(x).$$

The inner sum is a sum over all characters of G_2 minus a sum over the characters of G_1, which equals $q^2[x = 1] - q[x \in \ker(G_2 \to G_1)]$. The kernel consists of the classes with $p_3 = 0$.

If $p_3(x) \neq 0$, then x contains no proper prime power: ϖ^4 and ϖ^2 have no odd-degree terms. So $\pi(x) = \Psi(x)/4 = q^2/4$. □

All four statements were confirmed for $q = 2, 4, 8, 16$ by the script `char2_d2_rigidity.py`. In characteristic 2 the explicit formula thus degenerates completely at $d = 2$. The $q^2 - q$ characters of exact level 2 all have the same ψ_4, so Ψ takes only three values: on the identity class, on the other classes with $p_3 = 0$, and on the classes with $p_3 \neq 0$. The Oppermann classes are exactly the classes with $p_3 = 0$ (Lemma 2.2), and the identity class B_0 is where the fourth powers absorb the entire Weil lower bound.

Results on the variance of Ψ over Legendre intervals (an average over f) bound the number of intervals that deviate by a given amount; they do not exclude a single empty interval.

For fixed degree and $q \to \infty$, Bank, Bary-Soroker and Rosenzweig [1] prove the following. For monic $P \in \mathbb{F}_q[t]$ of degree k and $1 \leq m < k$, the number N of irreducibles in $\{P + h : \deg h \leq m\}$

satisfies

$$|N - q^{m+1}/k| \le c(k)\, q^{m+1/2},$$

provided that $m \ge 2$ if $\gcd(q, k(k-1)) > 1$, and $m \ge 3$ if q is even and $\deg P' \le 1$. They also note that the estimate fails outside this range.

Our blocks $B_e = \{t^4 + et^2 + h : \deg h \le 1\}$ have $k = 4$ and $m = 1$, and in characteristic 2 the centre $P = t^4 + et^2$ has $P' = 0$, so both conditions exclude them. Theorem 1.1(B) gives the exact count at this excluded point and shows how the asymptotic fails:

- for $q = 4^j$ the block $e = 0$ contains no irreducible, while the expected count is $q^2/4$;
- for $q = 2^{2j+1}$ the block $e = 0$ contains $q(q-1)/2$, about twice the expected count.

The blocks $e \neq 0$, with $q^2/4$ and $q(q-2)/4$ irreducibles, do agree with the asymptotic. The obstruction comes from the additive structure of $t^4 + et^2 + \alpha t$ (Proposition 3.1). For $m = 1$, $k = 4$ and q even, the conditions of [1] therefore cannot be dropped. By contrast, when $3 \mid q$, a case that is also excluded by $\gcd(q, k(k-1)) > 1$ at $k = 4$, $m = 1$, Theorem 1.4 shows that the asymptotic $q^2/4 + O(q)$ continues to hold without exception.

Theorem 1.1(C) has $q = 2$ fixed, so it lies outside the $q \to \infty$ regime altogether.

In the other direction, [1] gives nonemptiness for large q in most cases. An Oppermann sub-interval $O_{f,c} = \{f^2 + ct^d + h : \deg h \le d-1\}$ has $k = 2d$ and $m = d-1$. Since $N \ge q^d/2d - c(2d)q^{d-1/2}$, the sub-interval is nonempty as soon as $q > (2d\, c(2d))^2$, provided the conditions of [1] hold. We do not make the constant $c(2d)$ explicit.

Corollary 6.2 (of [1])**.** *For each d there is $q_0(d)$ such that every Oppermann sub-interval of degree $2d$ over $\mathbb{F}_q$, with $q \ge q_0(d)$, contains an irreducible polynomial in each of the following cases:*

(1) *q odd and $d \ge 3$;*
(2) *q odd, $d = 2$ and $\gcd(q, 6) = 1$;*
(3) *q even and $d \ge 4$;*
(4) *q even, $d = 3$, and the class has $p_3 \neq 0$.*

Proof. We check the conditions of [1] with $k = 2d$ and $m = d-1$; recall that $1 \le m < k$ is required.

- *q odd.* The only condition is $m \ge 2$ when $p \mid 2d(2d-1)$. It holds for $d \ge 3$. For $d = 2$, $k(k-1) = 12$, so it holds when $\gcd(q, 6) = 1$.
- *q even, $d \ge 4$.* Here $m \ge 3$, so both conditions hold.
- *q even, $d = 3$.* By Lemma 2.2 every member P of an Oppermann class has $p_5 = 0$. In characteristic 2 this gives $P' = p_3 t^2 + p_1$, so $\deg P' = 2$ when $p_3 \neq 0$, and the condition "$m \ge 3$ if $\deg P' \le 1$" does not apply.

□

The cases left open by the corollary are the following. For $d = 1$ the interval is too short for [1], but Theorem 1.1(A) and, for odd q, the count $(q-1)/2$ of irreducible $t^2 + bt + s_0$ settle it. For $d = 2$ the corollary is superseded: Theorems 1.1(B) and 1.4 give exact counts for every q, including $3 \mid q$, with no threshold. For q even with $d = 3$, the corollary is superseded as well. Theorem 6.4 in §6.1 counts all q^2 classes exactly for every q, including the q classes with $p_3 = 0$ (prefixes $(p_5, p_4, p_3) = (0, x, 0)$) that [1] misses. What remains is $q < q_0(d)$ for q odd with $d \ge 3$, and for q even with $d \ge 4$; these ranges are covered only by computation.

6.1. **All classes for $d = 3$ in characteristic 2: a field-tower count.** Let $q = 2^k$. By Lemma 2.2, the q^2 Oppermann classes for $d = 3$ are

$$C_{e,c} = \{t^6 + et^4 + ct^3 + s_2 t^2 + s_1 t + s_0 : \ s_2, s_1, s_0 \in \mathbb{F}_q\}, \qquad e, c \in \mathbb{F}_q.$$

Write $C_e = C_{e,0}$. Every member of $C_{e,0}$ has $P' = s_1$, so [1] does not apply to these classes. The classes with $c \neq 0$ are covered by [1] for large q, but the count below makes this unnecessary.

Proposition 6.3. (1) *For each c, the number $\pi(C_{e,c})$ does not depend on e.*
(2) *$E(C_{e,0}) = q^2$, so $\Psi(C_{e,0}) = 6\pi(C_{e,0}) + q^2$. For $c \neq 0$, $E(C_{e,c}) = 0$.*

Proof. (1) In characteristic 2 we have $(t+a)^6 = (t^4+a^4)(t^2+a^2) = t^6 + a^2t^4 + a^4t^2 + a^6$ and $(t+a)^4 = t^4 + a^4$. Neither has a t^3 term. On the other hand, $c(t+a)^3 = ct^3 + cat^2 + ca^2t + ca^3$. So $t \mapsto t+a$ maps $C_{e,c}$ bijectively onto $C_{e+a^2,c}$ and preserves irreducibility. Since $a \mapsto a^2$ is bijective, the claim follows.

(2) We count the proper prime powers of degree 6 that lie in some $C_{e,c}$. Sixth powers of linear polynomials and squares of cubics contain only even powers of t. Cubes of irreducible quadratics never lie in any $C_{e,c}$, as shown below. So no such prime power has a t^3 term, and $E(C_{e,c}) = 0$ for $c \neq 0$. Now take $c = 0$ and, by (1), $e = 0$.

- *Sixth powers.* $(t+a)^6 \in C_0$ only for $a = 0$; this is t^6, with $\Lambda = 1$.
- *Cubes.* For a quadratic $\varpi = t^2 + bt + c$, the coefficient of t^5 in ϖ^3 is b. So $\varpi^3 \in C_0$ forces $b = 0$, but then $\varpi = (t + \sqrt{c})^2$ is not irreducible.
- *Squares.* For a cubic $\varpi = t^3 + a_2t^2 + a_1t + a_0$ we have $\varpi^2 = t^6 + a_2^2t^4 + a_1^2t^2 + a_0^2$, which lies in C_0 exactly when $a_2 = 0$. Translation changes a_2 by $3c = c$, so the $(q^3-q)/3$ irreducible cubics are equidistributed over the values of a_2, and $(q^2-1)/3$ of them have $a_2 = 0$. Each contributes $\Lambda = 3$.

Hence $E(C_0) = 1 + (q^2-1) = q^2$. The cubes are excluded from every $C_{e,c}$, not only from C_0, since all of these classes have $p_5 = 0$. □

Theorem 6.4. *Let $q = 2^k$ and $e, c \in \mathbb{F}_q$.*

(1) *If k is odd, then $\pi(C_{e,c}) = q(q^2-1)/6$ for all e, c.*
(2) *If k is even, then*

$$\pi(C_{e,c}) = \frac{q(q-1)^2}{6} + q\,m(c), \qquad m(c) = \#\{\nu \in \mathbb{F}_q^\times \setminus (\mathbb{F}_q^\times)^3 : \ \mathrm{Tr}(c/\nu) = 1\}.$$

Here $m(0) = 0$, and $(q+2)/6 \leq m(c) \leq q/2$ for $c \neq 0$.

In particular, every Oppermann sub-interval with $d = 3$ in characteristic 2 contains at least $q(q-1)^2/6$ irreducibles if k is even and $q(q^2-1)/6$ if k is odd; this is positive for every q. For $c = 0$, $\Psi(C_e) = q^3 - (-1)^k(q^2-q)$.

Proof. Let $K = \mathbb{F}_q \subset E = \mathbb{F}_{q^3} \subset L = \mathbb{F}_{q^6}$. Fix c. By Proposition 6.3(1), $6q\,\pi(C_{e,c})$ is the number of $x \in L$ of degree 6 over K whose characteristic polynomial $t^6 + e_1t^5 + e_2t^4 + e_3t^3 + \dots$ has $e_1 = 0$ and $e_3 = c$.

Step 1 (conditions on x). With $S_m = \mathrm{Tr}_{L/K}(x^m)$, Newton's identity in characteristic 2 reads $S_3 = e_1S_2 + e_2S_1 + e_3$. So for $e_1 = S_1 = 0$ we get $e_3 = S_3$. The conditions are therefore $\mathrm{Tr}_{L/K}(x) = 0$ and $\mathrm{Tr}_{L/K}(x^3) = c$.

Step 2 (the relative trace u). Put $u = \mathrm{Tr}_{L/E}(x) = x + x^{q^3} \in E$; then $\mathrm{Tr}_{L/K}(x) = \mathrm{Tr}_{E/K}(u)$. Assume $\mathrm{Tr}_{E/K}(u) = 0$. We claim that x has degree 6 if and only if $u \neq 0$.

- If $u = 0$, then $x^{q^3} = x$, so $x \in E$.
- If $x \in \mathbb{F}_{q^2}$, then $x^{q^3} = x^q$, so $u = x + x^q \in K$, and $0 = \mathrm{Tr}_{E/K}(u) = 3u = u$.

Since the proper subfields of L are contained in E or in $\mathbb{F}_{q^2}$, this proves the claim.

Step 3 (the fibre over u). Fix $u \in E^\times$ with $\mathrm{Tr}_{E/K}(u) = 0$. The q^3 elements x with $\mathrm{Tr}_{L/E}(x) = u$ are $x = u\xi$ with $\xi + \xi^{q^3} = 1$. For these,

$$\mathrm{Tr}_{L/E}(\xi^3) = \xi^3 + (\xi+1)^3 = \xi^2 + \xi + 1.$$

Put $\alpha = \xi^2 + \xi$; then $\alpha^{q^3} = \alpha$, so $\alpha \in E$. As ξ runs over the fibre, α runs 2-to-1 over $\{\alpha \in E : \mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = 1\}$: the equation $\xi^2 + \xi = \alpha$ has its roots outside E exactly when $\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = 1$, and then the two roots are ξ and $\xi + 1 = \xi^{q^3}$. Finally,

$$\mathrm{Tr}_{L/K}(x^3) = \mathrm{Tr}_{E/K}(u^3(\alpha+1)).$$

Step 4 (counting in the fibre). The condition $\mathrm{Tr}_{L/K}(x^3) = c$ says that α lies in the affine K-hyperplane

$$H_{u,c} = \{\alpha \in E : \mathrm{Tr}_{E/K}(u^3\alpha) = \mathrm{Tr}_{E/K}(u^3) + c\}, \qquad |H_{u,c}| = q^2.$$

TABLE 4. The classes C_e ($d = 3$, $p_3 = 0$) in characteristic 2: exhaustive irreducible counts, in agreement with Theorem 6.4. For $q \le 16$ all e were counted; for larger q, $e = 0$ and, up to $q = 64$, $e = 1$. All q^2 classes $C_{e,c}$ agree with Theorem 6.4 for every $q \le 256$ (script `char2_d3_all.py`); for k even the values of $m(c)$ for $c \neq 0$ are $\{1,2\}$, $\{4,6\}$, $\{20,24\}$, $\{80,88\}$ at $q = 4, 16, 64, 256$.

q	$\pi(C_e)$	Theorem 6.4	$q^3/6$
2	1	$q(q^2-1)/6 = 1$	1.3
4	6	$q(q-1)^2/6 = 6$	10.7
8	84	84	85.3
16	600	600	682.7
32	5 456	5 456	5 461.3
64	42 336	42 336	43 690.7
128	349 504	349 504	349 525.3
256	2 774 400	2 774 400	2 796 202.7

The $\mathbb{F}_2$-linear functionals on E that vanish on the direction space of $H_{u,c}$ are $\alpha \mapsto \mathrm{Tr}_{E/\mathbb{F}_2}(\lambda u^3 \alpha)$ with $\lambda \in K$. Hence $\mathrm{Tr}_{E/\mathbb{F}_2}$ is constant on $H_{u,c}$ if and only if $u^3 \in K^\times$.

- If $u^3 \notin K^\times$, then $\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = 1$ on exactly $q^2/2$ points of $H_{u,c}$. This gives q^2 elements x, whatever c is.
- If $u^3 = \nu \in K^\times$, then on $H_{u,c}$ we have $\nu\,\mathrm{Tr}_{E/K}(\alpha) = \nu\,\mathrm{Tr}_{E/K}(1) + c = \nu + c$. So $\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = \mathrm{Tr}_{K/\mathbb{F}_2}(1 + c/\nu)$ throughout $H_{u,c}$. This gives $2q^2$ elements x if that value is 1, and none otherwise.

Step 5 (summing over u). There are $q^2 - 1$ admissible u.

- *k odd.* Here $3 \nmid q^3 - 1$, so cubing is bijective on $E^\times$ and on $K^\times$, and $u^3 \in K^\times$ forces $u \in K^\times$. But every $u \in K^\times$ has $\mathrm{Tr}_{E/K}(u) = 3u = u \neq 0$, so it is not admissible. All $q^2 - 1$ admissible u therefore contribute q^2, for every c, and $6q\,\pi = (q^2-1)q^2$.
- *k even.* Here $3 \mid q-1$, $3 \mid q^2+q+1$ and $\mathrm{Tr}_{K/\mathbb{F}_2}(1) = 0$. We sort the u with $u^3 = \nu \in K^\times$ by ν.
  - If $\nu \in (K^\times)^3$, its three cube roots lie in $K^\times$. They are not admissible, since $\mathrm{Tr}_{E/K}(u) = u \neq 0$.
  - If $\nu \notin (K^\times)^3$, its three cube roots lie in $E \setminus K$, because $K^\times \subset (E^\times)^3$. They have minimal polynomial $X^3 - \nu$, hence trace 0, and are admissible. There are $2(q-1)$ of them, and each contributes $2q^2$ when $\mathrm{Tr}_{K/\mathbb{F}_2}(c/\nu) = 1$ and nothing otherwise.
  - The remaining $(q^2-1) - 2(q-1) = (q-1)^2$ admissible u contribute q^2 each.

  So $6q\,\pi = (q-1)^2q^2 + 3 \cdot 2q^2\,m(c)$, which is the formula claimed.

  For the bounds on $m(c)$ when $c \neq 0$: as ν runs over the non-cubes, c/ν runs over $K^\times$ minus the coset $c(K^\times)^3$, which has $(q-1)/3$ elements. Since $K^\times$ contains $q/2$ elements of trace 1, we get $m(c) = q/2 - t(c)$ with $t(c) = \#\{y \in c(K^\times)^3 : \mathrm{Tr}(y) = 1\} \in [0, (q-1)/3]$.

The formula for $\Psi(C_e)$ follows from Proposition 6.3(2). □

Remark 6.5. The error term $\pi(C_e) - q^3/6$ is $-q/6$ for k odd and $-q^2/3 + q/6$ for k even. This is far smaller than the $O(q^{5/2})$ a monodromy argument would give.

It reflects a supersingularity that we observed numerically: for $q \le 16$, every character of exact level 2 or 3 of G_3 has an L-function whose inverse zeros satisfy $\gamma^6 \in q^3\{\pm 1, \pm i\}$. Figure 2 (right) shows these zeros for $q = 8$ next to the odd case $q = 7$. At level 2 this is a theorem: Proposition 6.1(3) gives $\gamma_\chi^4 = (-1)^k q^2$ for every character of exact level 2, which implies $\gamma_\chi^6 \in \{\pm q^3, \pm i q^3\}$. At level 3 it remains an observation. The proof above avoids these L-functions altogether. Summing over e replaces the level-3 condition by the two trace conditions $\mathrm{Tr}_{L/K}(x) = 0$ and $\mathrm{Tr}_{L/K}(x^3) = c$, which the tower $K \subset E \subset L$ makes linear.

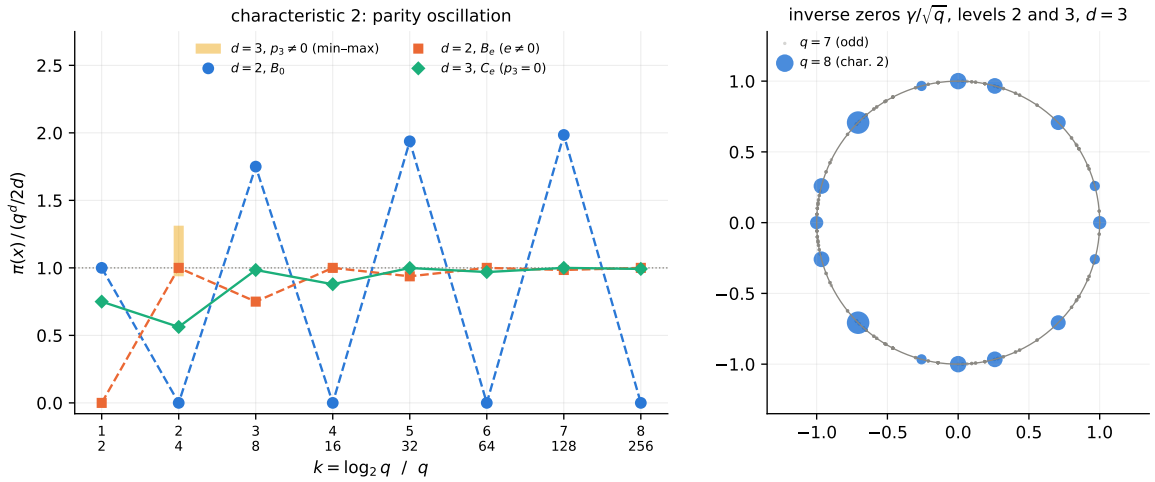


FIGURE 2. Characteristic-2 rigidity. *Left:* normalised counts $\pi(x)/(q^d/2d)$ for $q = 2^k$, $k \le 8$. Exact values from the Hayes FFT are shown as markers; the formulas of Theorem 1.1(B) and Theorem 6.4 are shown as lines. The bars give the range over the $d = 3$ Oppermann classes with $p_3 \neq 0$. For odd k this range reduces to the single value of C_e, as Theorem 6.4(1) predicts. For even k its endpoints are given by the two values of $m(c)$. *Right:* normalised inverse zeros $\gamma/\sqrt{q}$ of $L(u,\chi)$ for the characters of exact level 2 and 3 of G_3. For $q = 8$ (blue; marker area proportional to multiplicity) they occupy only 16 of the 24-th roots of unity. For $q = 7$ (grey; 630 zeros) they spread over the whole circle.

7. Questions

(1) In characteristic 2, are (A)–(C) the only empty Oppermann sub-intervals for all q and d? By Corollary 6.2, for each $d \ge 4$ only finitely many q can carry further empty sub-intervals. For $d \le 3$ the question is settled for every q: Theorems 1.1 and 6.4 give exact counts. So in characteristic 2 the only possible further empty sub-intervals have $d \ge 4$ and $q < q_0(d)$.

(2) For odd q, is every Oppermann sub-interval nonempty, for all q and d? Lemma 2.1 turns this into a statement about all short intervals of length $q^{n/2}$. By Corollary 6.2, for each $d \ge 3$ it fails for at most finitely many q. For $d = 2$ it holds for every odd q (Theorem 1.4). An explicit $q_0(d)$ combined with Theorem 1.2 would settle further small d completely. The trace-form method of Section 4 would need, for $d = 3$, the joint distribution of $\mathrm{Tr}(x^2)$ and $\mathrm{Tr}(x^3)$ on the subspace $\ker \mathrm{Tr} \subset \mathbb{F}_{q^6}$; it does not reduce to a quadratic form.

(3) Outside the characteristic-2 cases above, is there a constant $c > 0$ with $\min_x \pi(x) \ge c\, q^d/2d$ for all large d?

Data and code

- `hayes_oppermann.py`: Theorems 1.2 and 1.3.
- `verify_pointwise_oppermann.py`: brute-force census, $q^{2d} \le 2^{26}$.
- `compare_bruteforce.py`: class-by-class cross-check.
- `make_tables.py`, `make_figure.py`: Tables 2, 3 and Figure 1.
- `char2_d2_blocks.py`, `char2_d2_theorem.py`: Theorem 1.1(B).
- `swan_parity.py`: Theorem 1.1(C) and Table 1.
- `odd_d2_formula.py`: brute-force check of Theorem 1.4.
- `char2_d3_p3zero.py`, `char2_d3_all.py`: Table 4 and Theorem 6.4.
- `char2_d2_rigidity.py`: Proposition 6.1.
- `make_figure2.py`: Figure 2.

- `hayes_fft_fq.py`, `fqlib.py`, `small_q_gf.py`: taken from [3]. `fqlib.py` was extended by moduli for $q = 32, \ldots, 256$.

All scripts are in `code/` and their outputs in `data/` of https://github.com/Ruqing1963/oppermann-function-fields; archived as DOI 10.5281/zenodo.23353949.

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